MIT Academy of Engineering
(An Autonomous Institute)

PRODUCT /PROJECT TITLE

SCREENMATE MODULAR CLAMP

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MECHANICAL PROTOTYPING

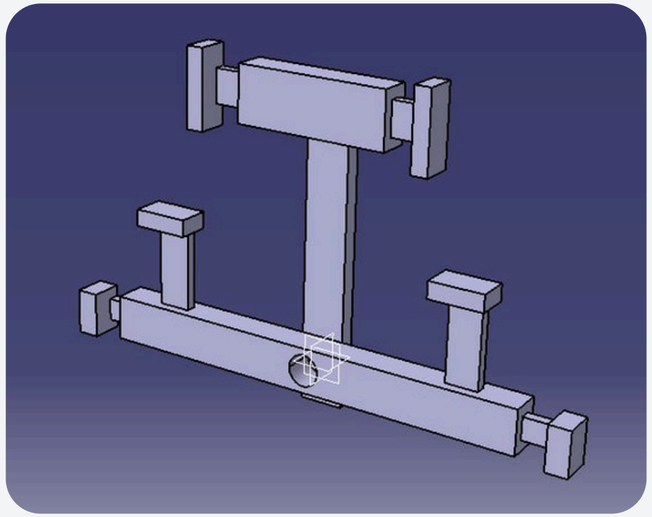
SCHOOL OF MECHANICAL ENGINEERING

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INTRODUCTION TO MECHANICAL PROTOTYPING

- **Our prototype:** Laptop + Mobile Holder with 360° rotation.
- **Purpose:** Check size, strength, and comfort of design before deployment.
- **Why 3D Printing:** Rapid iteration, low-cost, customizable geometry.
- **Design Goals:** Adjustable screen clamps, mobile holder, 360° pivot joints.



TECHNIQUES OF RAPID PROTOTYPING

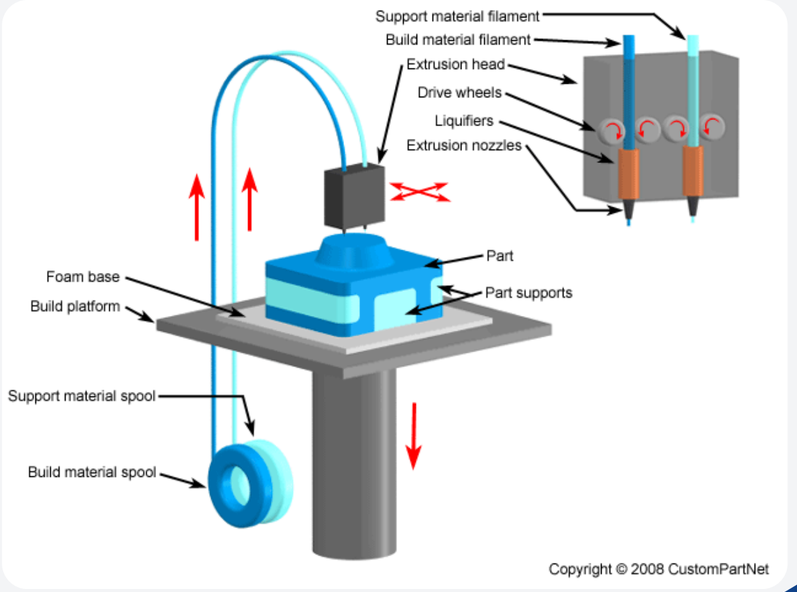
- **FDM (Fused Deposition Modeling)** – Using in our project
- Material: PLA (lightweight, easy to print).

Other Techniques (Possible Alternatives):

- SLS (Selective Laser Sintering): For stronger nylon-based pivots.
- CNC Machining: For metallic pivots instead of PLA.

Advantages of FDM for Our Prototype:

- Low-cost and quick.
- Easy to redesign adjustable parts (B, F).
- Suitable for functional testing with laptops and mobiles.



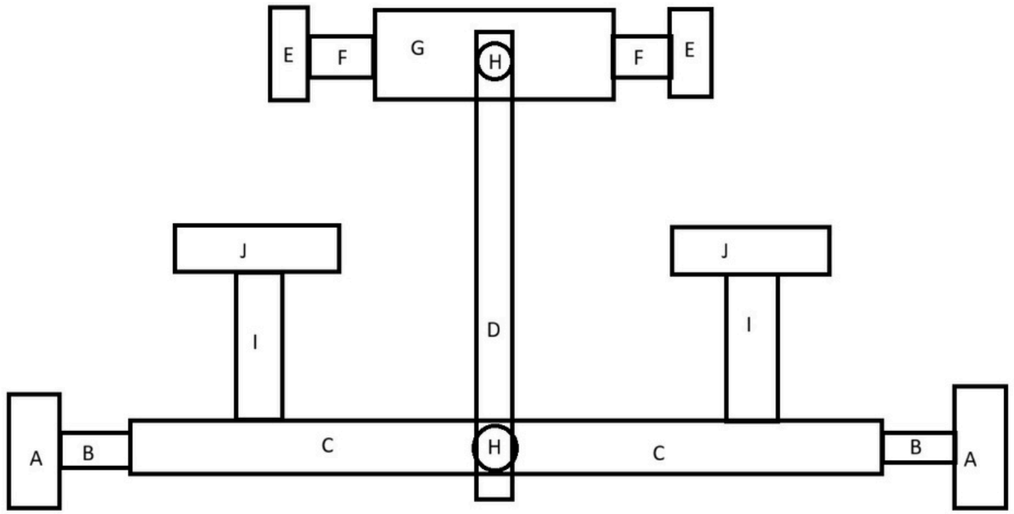
CONCEPT PRESENTATION

Geometry Highlights:

- **Screen Clamp Assembly:** Parts A, B, C, I, J
  - B is adjustable for screen width.
  - I and J grip the top bezel.
  - C is main bar.

- **Mobile Holder Assembly:** Parts E, F, G
  - F is adjustable for phone width.
  - G is central support; E is side brace.

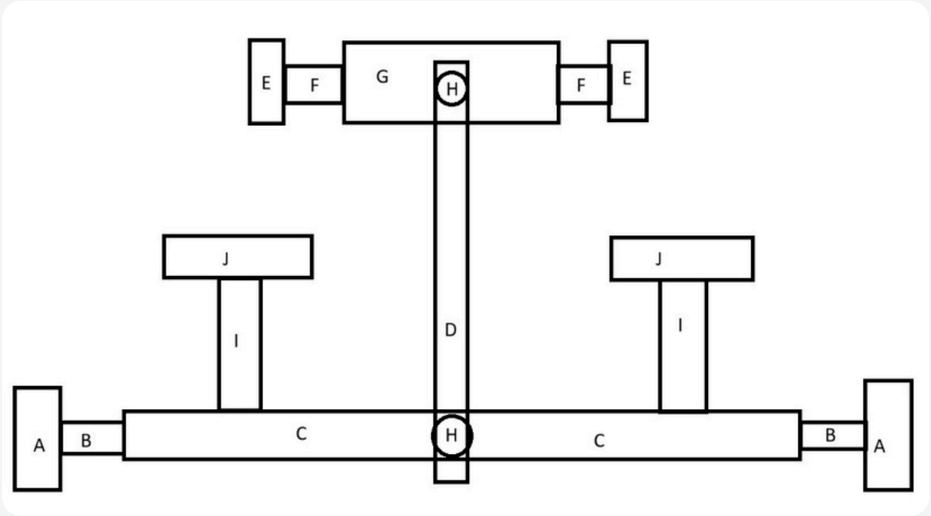
- **Central Spine D:** Connects assemblies.
- Pivot Points H: 360° rotatable joints.



CAD MODELING (CATIA)

Modeling Strategy:

- Parametric dimensions for screen/mobile sizes.
- Separate CATParts for modularity.



Key Dimensions:

- D: 220 × 20 × 10 mm (central spine)
- C: 88 × 12 × 10 mm (horizontal rails)
- B: Slider travel 20 mm
- F: Mobile clamp opening 55–95 mm
- H: Ø6 mm pivot pins

Assembly Notes:

- Use constraints for hinge rotation.
- Add fillets and nut pockets

METHODOLOGY

1. CAD Software used
- CATIA V5 — modify supplied 3D model, create parametric CATParts for changed components (A–J, D, H).
2. Slicing Software used
- Ultimaker Cura — import revised STLs, set Ultimaker 2+ / Ender 3 Pro profile, generate G-code.
3. 3D Printer used
- Creality Ender 3 Pro (FDM) — primary printer for prototypes; use Ultimaker 2+ settings if testing on that machine.
4. Rapid Prototyping Technique
- Fused Deposition Modeling (FDM) with PLA (TPU for pads) — print layer by layer, print test coupons (hinge, slider, clamp) before full parts.

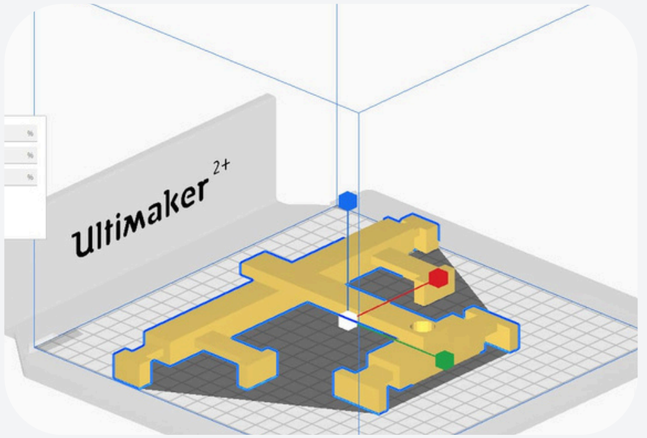
PRE-PROCESSING, PROCESS, POST-PROCESSING

Pre-Processing:

- Export STL with 0.1 mm tolerance.
- Check for non-manifold edges.

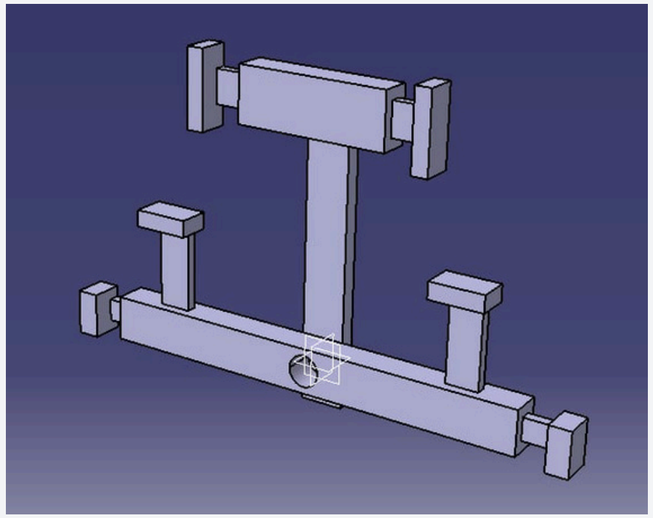
Cura Settings (Ultimaker 2+):

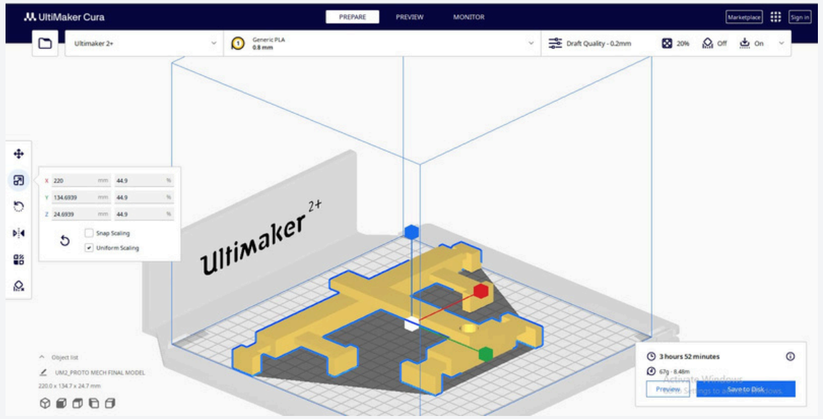
- Layer height: 0.2 mm
- Material: Generic PLA
- Infill: 40% gyroid
- Supports: Enabled for overhangs
- Estimated time: 3h 25m
- Material: 20g, 6.81m

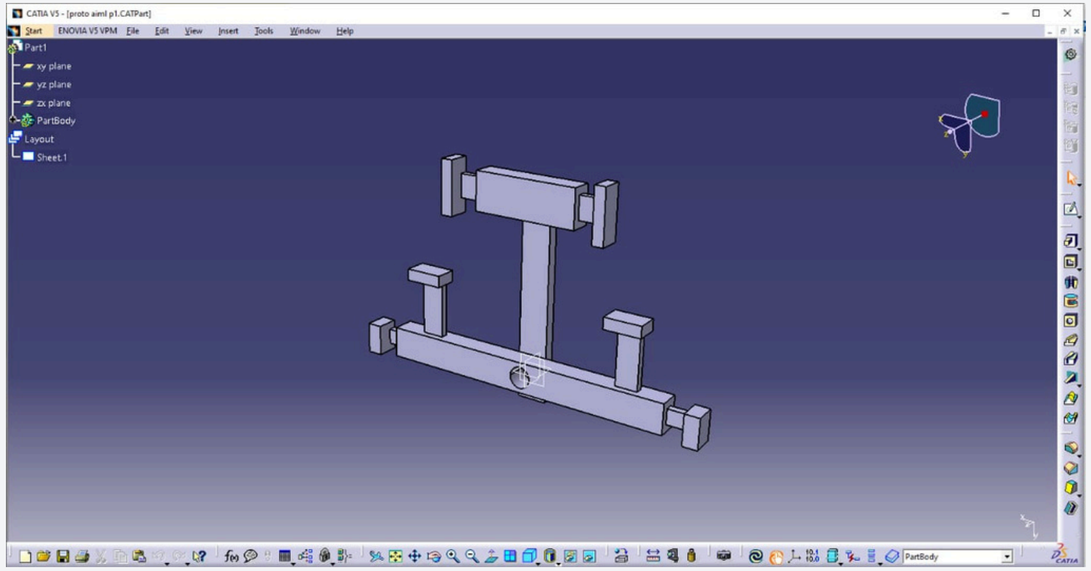


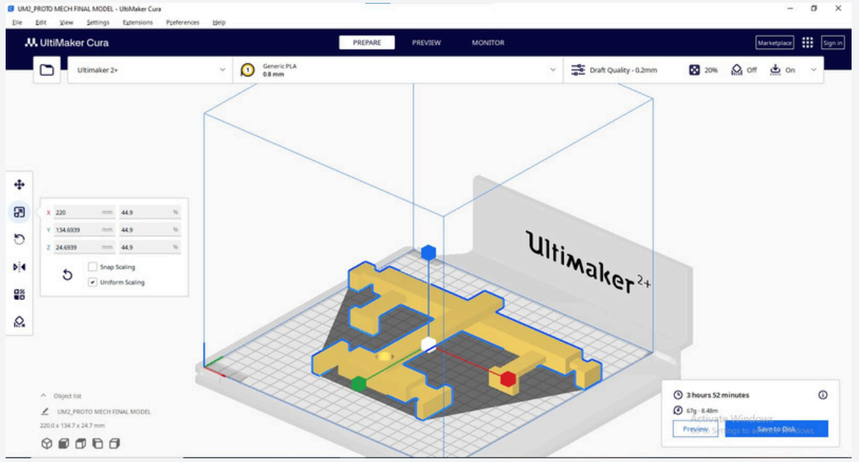
Post-Processing:

- Remove supports, sand mating faces.
- Insert TPU pads, assemble pivots.




UltiMaker Cura
PREPARE
PREVIEW
MONITOR
Ultimaker 2+
Draft Quality - 0.2mm
Ultimaker 2+
3 hours 52 minutes
Preview
Save to Disk


CATIA V5 - [proto aiml p1.CATPart]
Start
ENOVIA V5 VPM
File
Edit
View
Insert
Tools
Window
Help
Part1
xy plane
yz plane
zx plane
PartBody
Layout
Sheet.1
PartBody


UM2_PROTO MECH FINAL MODEL - UltiMaker Cura
UltiMaker Cura
PREPARE
PREVIEW
MONITOR
Ultimaker 2+
Draft Quality - 0.2mm
Ultimaker 2+
3 hours 52 minutes
Preview
Save to Disk

MATERIALS USED

Material	Use Case	Notes
PLA	Main structure	Easy to print, low warping
PETG	Optional for strength	Tougher, flexible
TPU	Grip pads	Soft, non-slip
Steel Pins	Pivot joints	Optional for durability

APPLICATIONS & GROUP ASSEMBLY

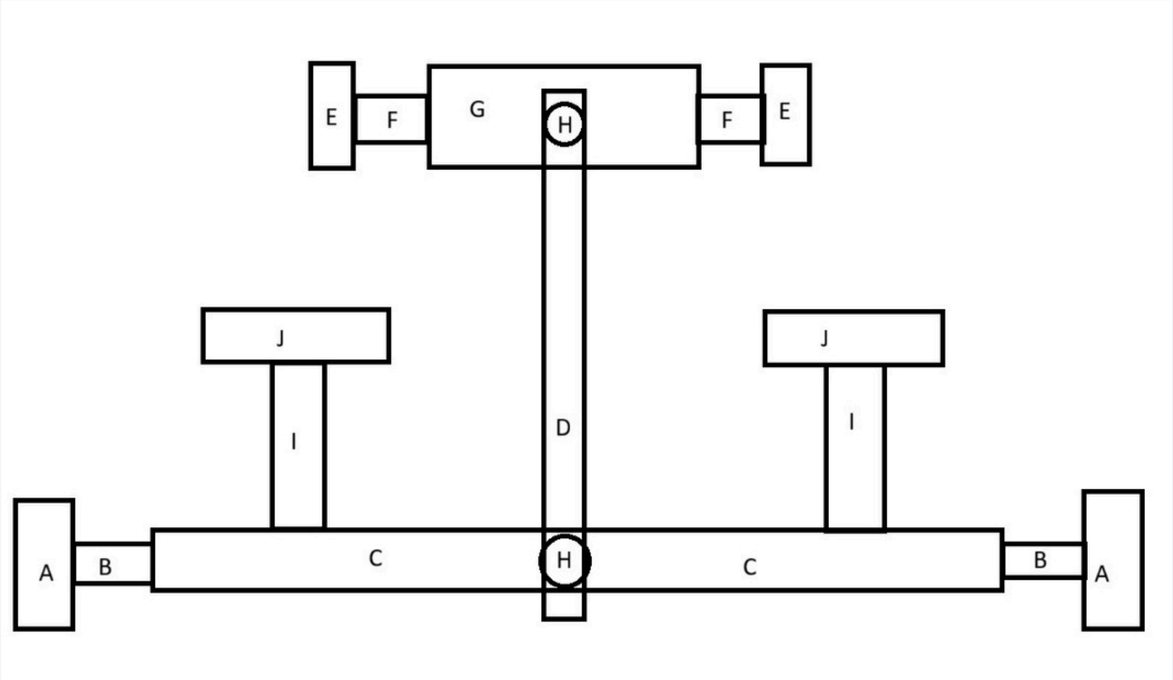
Where ScreenMate Modular Clamp Can Be Used:

1. **Students & Learners** – Hold phone and laptop together for online classes, note-taking, or dual-screen study.
2. **Office & Work-from-Home** – Keep phone at eye level while working on laptop, useful for multitasking and meetings.
3. **Presenters & Educators** – Use phone as a secondary camera for recording, demonstrations, or live streaming.

Team Roles & Contributions:

- **Arjun and om (CAD Designer): Designed all parts (clamps, spine, mobile holder) in CATIA with accurate dimensions.**
- **Krishna and Manmath(Slicing Operator): Converted CAD files into STL, set Cura slicing parameters, and optimized print quality.**

BILL OF MATERIALS



Part	Qty	Material	Total Cost (₹)
Central Spine (D)	1	PLA	25
Horizontal Rails (C)	2	PLA	40
Clamp Blocks (A, B)	4	PLA	66
Vertical Grips (I, J)	4	PLA + TPU	64
Mobile Holder (E, F, G)	3	PLA	56
Pivot Pins (H)	4	Steel/PLA	40
Screws (M3, M4)	8	Steel	60
Screws (M3, M4)	4	TPU	32

TOTAL ESTIMATED COST: ₹400

APPLICATIONS, ADVANTAGES, LIMITATIONS

- **Applications:**
  Remote learning, presentations, dual-screen workflows.

- **Advantages:**
  - Modular, adjustable, portable.
  - 360° rotation.
  - Low-cost prototyping.

- **Limitations**:
  - PLA may deform under heat.
  - Not suitable for heavy tablets.
  - Requires precise tolerances for fit.

THANK YOU